Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH

B. Müller Alexander Osika Kai Xue Ben Cassie Maarten Antonie Visser

July 5, 2026

Paper release: `r1512` **Released:** July 5, 2026

Abstract

OPH needs a finite carrier for screen cuts. Observers see finite pieces of the screen, and this model turns those pieces into patch bodies with ports, records, repair channels, and synchronization rules.

The construction gives a twelve-port screen sieve, edge weights used by later papers, central records with Born–Lüders readout, Bell/CHSH records with the Tsirelson bound, and checkpoint restoration for observer continuation after repair. It also states the public evidence rule: hardware runs matter only when the device, calibration, circuit, data, and independent check are public.

What This Paper Contributes

The rest of the OPH stack talks about observer patches, records, repairs, and screen cuts. This paper gives those words a finite carrier. It separates the abstract observer patch from its geometric support chart and from any physical or digital implementation. That distinction is what prevents the theory from depending on a particular piece of hardware or a convenient drawing of a sphere.

The concrete contribution is the regulated screen architecture. Echosahedral carriers supply a twelve-port reference interface, edge sectors supply the heat-kernel/Casimir weights used by the quantitative branch, central records give the Born–Lüders event surface, Bell/CHSH records give the Tsirelson bound, and checkpoint restoration states what it means for an observer to continue after repair. Hardware claims are kept behind a public evidence rule: manifests, hashes, calibration records, raw or reduced traces, and exact verifier receipts carry the weight.

Contents

1 Scope

OPH microphysics uses finite observer patches with echosahedral interfaces. A spherical screen is a geometry chart for observer-visible cuts. The cellulated-sphere gauge-register construction is a regulator and digital calibration chart. The premise is:

> Observers access finite observer-visible cuts. A spherical screen is a geometry chart for such cuts. The underlying fixed-cutoff implementation surface is a federation of finite overlap-facing observer patches, with echosahedral local interfaces and recurrent toroidal subchannels.

The paper's contract is:

1. state the fixed-cutoff theorem exports used by the other OPH papers;
2. use federated patch carriers as the microphysical architecture;
3. keep mathematical theorem surfaces separate from public hardware evidence surfaces;
4. keep the octahedral $\mathbb{Z}_2/S_3$ finite-group model at appendix-level calibration scope;
5. require a public hardware evidence protocol before hardware claims receive paper weight.

2 Canonical Observer-Patch Semantics

This paper owns the fixed-cutoff observer vocabulary used by the rest of the OPH stack. An observer patch is first an operational algebraic object:

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i is the accessible state, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the exact central record algebra or a declared approximate record algebra, $\mathcal{I}_e$ is an overlap-visible interface algebra, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the visible restriction map, $\mathcal{U}_i$ is the allowed family of update and repair instruments, and Chk_i is checkpoint data sufficient for the declared continuation problem.

Definition 2.1 (Abstract observer patch)**.** *An abstract observer patch is the tuple O_i above, considered up to isomorphism of accessible algebras, record statistics, visible restrictions, repair instruments, and checkpoint continuation.*

Definition 2.2 (Support patch)**.** *A support patch is a geometric chart for an abstract observer patch on a branch with reconstructed geometry. Examples are a cap $P_i \subset S^2$, a collar, or a causal diamond. The support patch supplies chart data for modular flow, entropy variation, and overlap comparison.*

Definition 2.3 (Carrier patch)**.** *A carrier patch is a physical or digital implementation satisfying*

$$\mathrm{Realize}_\varepsilon(H_i) \longrightarrow \mathsf{O}_i,$$

meaning that the implementation induces the declared accessible algebra, visible interface statistics, record readout, repair instruments, and checkpoint continuation within error ε.

The echosahedral body used below is a reference carrier architecture. It is a finite implementation surface for twelve-port interfaces, records, repair channels, and recurrent subchannels. The observer identity lives in the quotient data exposed by O_i, so a change of hidden coordinates, port labels, or substrate presentation is physically silent when it preserves the visible interface and record process.

The tuple O_i does not impose equal primitive observer size. Fixed-cutoff OPH requires bounded accessible algebras and declared interfaces; it also permits different connected support regions and different finite or regulated algebras. In the carrier picture, an observer-supporting object may be a variable finite federation of screen cells. A theorem selecting isomorphic fixed-capacity elementary carriers would need extra hypotheses: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlaps, refinement preservation of the cell type, and a proof that one cell or a fixed block of cells is the primitive carrier.

Claim 2.4 (Implementation-invariance target)**.** *Two carrier patches that induce isomorphic visible interface algebras, record statistics, repair maps, and checkpoint continuation are physically equivalent for all OPH observables in the declared quotient.*

This claim is theorem-level only on a branch that supplies the required quotient isomorphism and error bound. It fails if a predicted OPH observable depends on hidden carrier coordinates after the visible algebra, record statistics, repair maps, and checkpoint data have been held fixed.

Definition 2.5 (Operational observer test)**.** *A candidate physical system realizes an observer patch when it supplies a bounded accessible interface, durable re-readable records, self-read or internal state estimation, record-conditioned future behavior, boundary prediction better than shuffled-record controls, and checkpoint continuation within a declared error.*

3 Claim Taxonomy

The paper uses five claim levels.

C1. Mathematical fixed-cutoff claims. These are finite-algebra statements about patches, overlaps, records, repair interfaces, event algebras, and checkpoint laws. They are allowed to be theorem-bearing. The generic overlap network is a finite constraint code only; QECC distance, min-cut resilience, spectral convergence, BFT liveness, and hardware speedup need their own certificates.

C2. Regulator-chart claims. These identify a convenient finite model, such as a spherical cellulation or a digital finite-group calibration chart, as a calculational chart. Their claim level is regulator geometry and calibration.

C3. Federated-carrier claims. These define a candidate implementation architecture: observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, and repair loops.

C4. Public hardware-evidence claims. These may be cited only when the raw or reduced evidence is present in a public OPH evidence bundle with stable hashes, manifests, calibration files, and exact-verifier receipts.

C5. Boundary statements. This paper leaves laboratory-hardware identification, unique UV completion, hardware solution of the Yang–Mills mass gap, computational complexity collapse, and first-principles hadron masses outside its support boundary.

4 The Public Evidence Rule

Hardware-facing language is admissible only under the following rule.

> Any hardware evidence cited as evidence in this paper must be represented by a public evidence bundle in the OPH repository, or by a public pinned subrepository commit, with enough raw data and metadata for an external reader to check the claim.

Private notebooks, local runtime logs, unpublished bench transcripts, and development-repo notes may motivate architecture choices, but they do not carry evidential weight inside the paper. A hardware evidence bundle must include, at minimum:

1. a manifest with stable bundle identifier, date, operator, body identifier, controller identifier, firmware hash, mesh/body hash, and wiring map hash;
2. raw readout files, such as coupling matrices, MDD or discharge-timing traces, dark baselines, low-power sweeps, ring-diversity scans, and calibration logs;
3. body and board provenance, including photographs or signed photo hashes where relevant;
4. the task definition, scorebook, repair or rerank law, and exact-verifier program;
5. exact-verifier receipts for any high-level task claim;
6. negative controls, shuffle or replay controls where applicable, and a statement of non-claims.

This rule prevents dependence on hidden laboratory state. The theorem sections use finite algebras, declared patch interfaces, and stated branch assumptions.

5 From Spherical Screens to Federated Patch Carriers

The spherical screen is useful because an observer-accessible cut often has an effective closed two-surface description. A cellulated sphere can encode finite capacity, caps, collars, edge centers, and observer-visible cuts. The microscopic carrier is the finite patch federation. The chart is the observer-facing presentation of its visible data.

The sphere also fixes the symmetry bridge used by the rest of OPH. Caps on S^2 are the geometric support regions whose modular flows become Lorentz boosts in the controlled scaling branch. The conformal group of S^2 is $\mathrm{SO}^+(3,1)$, so the same observer-facing chart supplies the kinematic bridge to emergent $3+1$D spacetime. Finite cellulations of the chart supply the regulator side: patch ports, edge sectors, collars, and overlap checks.

At fixed cutoff, the fundamental object is a finite patch federation. Each patch has:

1. an internal finite state algebra;
2. a bounded family of exposed overlap ports;
3. a local record algebra;
4. a readout map from internal state to port-visible packets;
5. a repair interface that changes local state in response to mismatch;
6. a checkpoint interface that exposes enough observer-accessible data to define continuation.

Definition 5.1 (Sphere fold)**.** *Fix a regulator r. Let Σ_r be the finite presentation space of a patch federation, let Γ_r be the presentation-redundancy groupoid, and let*

$$Q_r = \Sigma_r/\Gamma_r$$

be the physical quotient. Let $\pi_r : \Sigma_r \to Q_r$ be the quotient map, let

$$n_r : Q_r \to Q_{r,\mathrm{nf}}$$

be the accepted repair normal-form map on the declared physical quotient, and let

$$\chi_{S,r} : Q_{r,\mathrm{nf}} \to \mathsf{Chart}_S(r)$$

be the support-visible spherical screen chart. Its outputs are caps, collars, cuts, edge-sector records, boundary data, and geometric readouts. The sphere fold of a finite presentation $s \in \Sigma_r$ is

$$\mathrm{Fold}_{S,r}(s) := \chi_{S,r}(n_r(\pi_r(s)))\,.$$

Two finite presentations have the same sphere fold exactly when their repaired quotient normal forms induce the same support-visible caps, collars, edge-sector records, and observer-facing screen readouts.

The fold adds no independent dynamics. Repair is the finite patch operation that compares overlaps, updates records, and reaches the accepted normal form. Folding names the spherical chart presentation of that repaired quotient state. In plain language, sphere folding is what overlap repair looks like on the observer-facing screen.

Definition 5.2 (Finite source-bridge readouts)**.** *For cosmology runs that attempt a first-principles primordial bridge, the carrier must also expose quotient-visible maps*

$$\begin{gathered}\mathsf{StressRead}_r,\quad \mathsf{ScalarSourceMap}_r,\quad \mathsf{ClockRead}_r,\quad \mathsf{RepairTangent}_r,\\ \mathsf{VolumeJacobianRead}_r,\quad \mathsf{ScreenMassMatrixRead}_r,\quad \mathsf{LowModeProjectorRead}_r,\\ \mathsf{ScalarPrecisionRead}_r,\quad \mathsf{QuotientMeasureRead}_r,\quad \mathsf{ScalarReleaseEnergyRead}_r,\\ \mathsf{DiamondRead}_r,\quad \mathsf{ProperScaleRead}_r,\quad \mathsf{ModularRemainderRead}_r,\quad \mathsf{StressMomentRead}_r,\\ \mathsf{FaceFluxRead}_r,\quad \mathsf{ReactionRead}_r,\quad \mathsf{TetradRead}_r,\quad \mathsf{ScalarRefinementRead}_{sr},\\ \mathsf{RadialWindowRead}_r,\quad \mathsf{SpatialGeometryRead}_r,\quad \mathsf{AreaVolumeRead}_r,\quad \mathsf{LapseShiftRead}_r,\\ \mathsf{LineageRead}_r,\quad \mathsf{ScreenEmbeddingRead}_r,\quad \mathsf{ModeFormRead}_r,\quad \mathsf{CosmoGeomRead}_r,\\ \mathsf{ProperTimeRead}_r,\quad \mathsf{ComovingMetricRead}_r,\quad \mathsf{SpatialModeRead}_r,\quad \mathsf{AngularProjectorRead}_r,\\ \mathsf{FreezeoutMapRead}_r,\quad \mathsf{FreezeoutSurfaceRead}_r.\end{gathered}$$

They are defined on the same settled-form carrier used for the geometric record packet and must factor through the physical quotient: hidden carrier coordinates, port relabelings, and accepted repair schedules cannot change their output. The finite packet must expose enough data to reconstruct, for every active sector,

$$T_I^{ab},\quad Q_I^a,\quad u_I^a,\quad \rho_I,\quad p_I,\quad \pi_I^{ab}.$$

The source-bridge maps are instrumentation for the screen-to-radial theorem. They do not turn a screen covariance into a primordial spectrum unless the source-stress, single-clock, repair-gap, freeze-out, geometric-screen-scalar, scalar-precision, source-release-energy, refinement-tilt, radial-prior, null-space, finite-window, and forward-residual receipts also pass.

Target 5.3 (Geometry-readout quotient factorization). *The geometry-facing readouts in Definition 5.2 must factor through the same quotient normal form as the source readouts. In particular, a finite carrier patch, graph vertex, implementation point, or worker-local shard is not one canonical OPH screen cell unless a separate measure-realization theorem proves that identification. A promotable evidence bundle must expose a geometry manifest, metric-form hashes, orientation and topology, boundary conditions, lineage maps, lapse and shift, source embedding, operator assembly, scale certificate, and refinement maps.*

A spherical screen is then a coarse chart over a federation of such patches. It is what a particular observer-facing support cut looks like after quotienting hidden implementation details and choosing a geometric presentation. The microscopic carrier may be graph-like, federated, multi-patch, and locally polyhedral.

The P, N closures supply the geometric side of that chart. If $P_\star = a_{\text{cell}}/\ell_\star^2$ and $N_{\text{CRC}} = A_{\text{screen}}/(4\ell_\star^2)$, then an equal-area cellulation has

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_{\text{CRC}}}{P_\star} \simeq 8.12 \times 10^{122}.$$

This counts canonical geometric screen cells. It does not assign an independent Hilbert factor to each cell. In particular $P_\star/4 \simeq 0.408$ nats is about 0.588 bits, so it is not by itself the logarithm of an integer-dimensional autonomous cell algebra. The area law lives on shared cuts, edge centers, and constrained overlap data.

Definition 5.4 (Federated patch carrier). *A fixed-cutoff federated patch carrier is a tuple*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}_{i\in V}, \{\mathcal{I}_e\}_{e\in E}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\}),$$

where V is a finite set of patches, E is a finite overlap graph, $\mathcal{A}_i$ is the local finite algebra of patch i, $\mathcal{I}_e$ is the finite interface algebra on overlap $e = \{i, j\}$, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the declared visible restriction, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the patch record algebra, and $\mathcal{U}_i$ is the allowed local update and repair interface.

The carrier is *observer-facing* when every physical claim is made through the visible restrictions $\pi_{i,e}$, record algebras $\mathcal{R}_i$, and the quotient-local observables specified by the OPH consensus package. Hidden coordinates may exist, but they are not directly physical.

5.1 Canonical multiresolution regulator chart

Definition 5.5 (Reference multiresolution carrier). *Choose the nested geodesic icosahedral subdivisions of S^2. A regulator is*

$$r = (m, L, b),$$

where m is the subdivision depth, L is the finite support or volume scale of the exported carrier, and b is a boundary/phase label. Every added cell, collar, edge-center, or observer-visible channel j carries a finite algebra

$$D_j = \bigoplus_{\alpha\in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j. At regulator r,

$$\widetilde{M}_r = \bigotimes_{j\in\mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j\in\mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit W_r maps these multiresolution coordinates to the declared patch, port, collar, and gauge coordinates:

$$M_r = \mathrm{Ad}(W_r)(\widetilde{M}_r), \qquad \widehat{\omega}_r = \widetilde{\omega}_r \circ \mathrm{Ad}(W_r^*).$$

For $r \preceq s$, refinement adds detail factors. The refinement embedding and coarse-graining are

$$\iota_{rs}(A) = W_s\big[(W_r^* A W_r) \otimes \mathbf{1}\big] W_s^*,$$

and

$$Q_{sr}(X) = W_r\big[(\mathrm{id} \otimes \tau_{s\setminus r})(W_s^* X W_s)\big] W_r^*.$$

The embedded map $E_{sr} = \iota_{rs} Q_{sr}$ is the canonical state-preserving conditional expectation.

Theorem 5.6 (Fixed-cutoff regulator certificate)**.** *The maps in Definition 5.5 are unital, completely positive, composition-compatible, and preserve the faithful reference states. The refinement embeddings are unital injective $*$-homomorphisms, the E_{sr} are faithful conditional expectations, and the reference modular groups restrict exactly along the tower. The inductive limit has canonical finite-regulator renormalization maps given by these conditional expectations; their martingale tails are the finite-volume and lattice-spacing Cauchy errors used by the compact and main papers.*

Proof. In bare coordinates all statements reduce to tensoring an observable with the identity and averaging the detail factors against the faithful product state. Composition follows from product-state associativity. Conjugating by the presentation circuits gives the physical patch coordinates without changing the algebraic identities. □

Remark 5.7 (Reference state, ground state, and vacuum)*.* The state $\widehat{\omega}_r$ is the finite-regulator reference state. It may be called a finite-volume ground state only after a positive self-adjoint transfer or Hamiltonian has been constructed and shown to have that state as its ground state. The term continuum vacuum is reserved for the GNS/OS limit after positive energy and the declared vacuum-sector uniqueness or superselection condition have been proved.

Remark 5.8 (Regulator promotion rule)*.* A finite evidence bundle implements this regulator branch only when it exports the factor list, presentation circuit, refinement maps, conditional-expectation identities, transported state errors, renormalized observable tails, and positive-transfer or reflected-Gram certificates. A finite cellulation, finite-group score, or stable numerical extrapolation by itself remains a regulator-chart result.

Definition 5.9 (Clocked spatial readout package)**.** *A finite cosmological run that claims FLRW spatial curvature must export the clock and spatial geometry readouts separately from screen/collar defect bookkeeping:*

$$\mathsf{SpatialMetricRead}_r, \quad \mathsf{SpatialFrameRead}_r, \quad \mathsf{SpatialConnectionRead}_r, \quad \mathsf{SpatialTransportRead}_r, \quad \mathsf{CurvatureHolonomy}$$

The transport readout uses the spatial Levi–Civita connection on the clock slice. Finite permutation holonomy of an S^2 screen triangulation or S_3 collar defect is a repair/sieve diagnostic unless these readouts identify it with a spatial Levi–Civita holonomy under quotient, transport, and refinement naturality.

6 The Echosahedral Patch Object

An echosahedral patch is the reference local body for this microphysics. The term is architectural: a bounded observer patch with a highly symmetric multi-port interface.

Definition 6.1 (Echosahedral patch)**.** *An echosahedral patch is a finite patch object*

$$\mathcal{E} = (\mathcal{A}, \{P_a\}_{a=0}^{11}, \{\rho_a\}_{a=0}^{11}, \mathcal{R}, \mathcal{M}, \mathcal{U}, \mathcal{G}_{\mathcal{E}}),$$

where:

1. $\mathcal{A}$ *is the internal finite algebra;*
2. $P_0, \ldots, P_{11}$ *are twelve labeled overlap ports;*
3. $\rho_a : \mathcal{A} \to \mathcal{I}_a$ *are port readout maps;*
4. $\mathcal{R} \subseteq Z(\mathcal{A})$ *is the observer-accessible record algebra;*
5. $\mathcal{M}$ *is a finite mismatch-score family on exposed port packets;*
6. $\mathcal{U}$ *is a finite family of local update and repair instruments;*
7. $\mathcal{G}_{\mathcal{E}}$ *is a declared finite symmetry or approximate-symmetry group of the port arrangement.*

Theorem 6.2 (Icosahedral screen-sieve theorem)**.** *Assume the fixed-cutoff OPH screen branch is represented by a quantum-link triangulation of S^2, with finite Hilbert spaces on links, Gauss constraints at vertices, boundary-gauge-invariant physical algebras, and a refinement-stable MaxEnt quotient-ensemble background that is locally six-valent away from curvature defects. Let the discrete curvature charge at a vertex be*

$$q_v = 6 - \deg(v).$$

If the defect cost is convex in the charges and the edge-center collar decomposition exposes each irreducible positive defect as one central port, then the invariant screen contraction is sampled by exactly twelve indistinguishable ports. Under the no-marked-point MaxEnt symmetry rule those ports form the 12*-vertex icosahedral orbit. Therefore an invariant scalar screen load X, in particular $X = \log(N/\pi)$, is read locally as $X/12$.*

Proof. For a triangulated sphere with V vertices, E edges, and F faces, Euler and triangular incidence give

$$V - E + F = 2, \qquad 3F = 2E, \qquad E = 3V - 6.$$

Therefore

$$\sum_v q_v = \sum_v (6 - \deg(v)) = 6V - 2E = 12.$$

This is the discrete Gauss–Bonnet charge of the spherical screen. A locally hexagonal reference background has $q_v = 0$ away from defects. With fixed total positive charge 12, a convex defect cost such as $\sum_v q_v^2$ is minimized by twelve unit defects $q_v = 1$, while tetrahedral and octahedral alternatives concentrate the same charge into four $q = 3$ defects or six $q = 2$ defects. The edge-center collar package turns each unit fivefold defect into a central readout algebra $Z(\mathcal{A}(B_i))$, so the defects are the ports. The no-marked-point MaxEnt rule makes the twelve ports indistinguishable. The maximal finite rotation symmetry of the sphere is the icosahedral group $I \simeq A_5$, and orbit-stabilizer gives

$$|I|/|C_5| = 60/5 = 12.$$

Any invariant scalar assigned to a transitive twelve-port orbit is split equally across the ports, so the local port read is $X/12$. □

The theorem fixes the canonical twelve-port branch. Its role is to give the reference patch enough boundary structure to support nontrivial overlap comparison, symmetry tests, verifier-shadow behavior, and cross-patch routing while remaining small enough for explicit evidence records. The theorem selects the exposed interface architecture. The dimension of the observer's internal algebra belongs to a separate carrier theorem over ports, records, repair maps, and refinement.

Remark 6.3. The word "echosahedral" is intentionally implementation-facing. In a mathematical theorem, one uses the tuple above. In a public hardware evidence bundle, one may instantiate the tuple using a specific twelve-port body, wiring map, controller, firmware hash, readout protocol, and evidence manifest.

7 Toroidal Recurrence Is Local

Toroidal hardware supplies a local recurrence and mode-competition surface. A toroidal subchannel is a recurrent internal path inside a patch or between a bounded set of ports. It can recycle boundary information, support settling dynamics, and expose winding-like or phase-locking observables.

Global scalar order parameters are often misleading in recurrent local systems. A toroidal or ring-like patch may fail to show global alignment while selecting a stable winding class, a local coherence island, or a persistent twisted state. The validation metrics for toroidal subchannels should include:

1. local coupling matrices, with total brightness treated as a secondary summary;
2. recurrence and ring-diversity statistics, with global response treated as a secondary summary;
3. winding-sensitive or phase-lock-sensitive summaries when phase data are available;
4. matched controls that distinguish chamber-mediated dynamics from host-side filtering;
5. exact-verifier receipts for task-level claims.

The OPH interpretation is simple: toroidal recurrence supplies local memory and mode competition inside the patch federation. Echosahedral symmetry supplies a reference and consensus geometry. The universe, at this level of description, is a federated repair system with many local carriers.

8 Patch Federation and Overlap Synchronization

Given a federation of echosahedral patches, overlaps are formed by routing ports into declared interface pairs or interface hyperedges. For an edge $e = \{(i, a), (j, b)\}$, the exposed packets are

$$x_{i,a} = \rho_{i,a}(s_i), \qquad x_{j,b} = \rho_{j,b}(s_j),$$

and the edge mismatch is a nonnegative function

$$\Phi_e(x_{i,a}, x_{j,b}) \geq 0.$$

The total visible mismatch is

$$\Phi(s) = \sum_{e \in E} \Phi_e\big(\rho_{i,a}(s_i), \rho_{j,b}(s_j)\big).$$

The local repair contract is the consensus-paper contract: accepted repairs do not increase the declared touched-overlap mismatch, and exact fixed-cutoff branches lower the relevant mismatch unless the local visible datum is repaired. The implementation is a bounded patch operation:

1. read the exposed overlap packets;
2. compare them through a declared commensurability map;
3. choose an allowed local update or rerank move;
4. write a record of the move;
5. expose the new port packet;
6. let the exact verifier decide any high-level task claim.

Proposition 8.1 (Federated synchronization contract)**.** *Suppose a finite patch federation has a declared mismatch functional Φ, a finite repair menu, and an accepted-repair rule such that every accepted repair lowers Φ unless the touched visible datum is locally repaired. Then every repair sequence terminates at a visible local normal form. If the union-collar gluing and repair-completeness hypotheses of the OPH consensus theorem hold on the quotient-local carrier, the terminal physical observable state is schedule-independent on that carrier. The carrier is thereby a finite constraint-code implementation surface; no QECC distance, min-cut formula, spectral mixing rate, or wall-clock liveness bound follows unless the corresponding extra certificate is supplied.*

Proof. The first sentence is the finite Lyapunov argument: Φ takes values in a finite ordered set of declared mismatch scores and strictly decreases on nontrivial accepted repairs. The second sentence is not reproved here; it is exactly the quotient-local confluence package supplied by the OPH consensus theorem. This paper supplies the federated carrier and visible interface on which that theorem is read. □

9 Records, Observers, and Checkpoints

An observer is not added as a metaphysical extra. In this paper an observer is an operational pattern in a patch or patch subfederation with persistent access to:

1. an observer-facing local algebra;
2. a record algebra;
3. a stable readout/update interface;
4. enough checkpoint data to define future observer-accessible probabilities.

Definition 9.1 (Federated observer checkpoint)**.** *For an observer-supporting patch subfederation O, a checkpoint at semantic cut D is*

$$\mathrm{Chk}_O(D) = \big(\mathcal{R}_O(D), \rho_O^{\mathrm{acc}}(D), \mathfrak{I}_O^{\mathrm{ext}}(D), \mathcal{L}_O^{\mathrm{sem}}(D), \mathfrak{B}_O(D)\big),$$

where $\mathcal{R}_O(D)$ is the observer-accessible record algebra, $\rho_O^{\mathrm{acc}}(D)$ is the state restricted to observer-accessible records and visible interfaces, $\mathfrak{I}_O^{\mathrm{ext}}(D)$ is the external port-interface tuple, $\mathcal{L}_O^{\mathrm{sem}}(D)$ is the semantic future-law class for the same physical instruments and event algebra, and $\mathfrak{B}_O(D)$ is the public or internal provenance bundle needed to replay the checkpoint at the declared accuracy. Worker IDs, queue positions, retry counters, repair-cycle indices, timestamps, and packet latencies live in $\mathfrak{B}_O(D)$ unless a branch declares them physical inputs; they are not observer identity or observer time.

Theorem 9.2 (Checkpoint continuation at fixed cutoff)**.** *If two federated observer checkpoints agree exactly on $\mathcal{R}_O(D)$, $\rho_O^{\mathrm{acc}}(D)$, $\mathfrak{I}_O^{\mathrm{ext}}(D)$, and $\mathcal{L}_O^{\mathrm{sem}}(D)$, then they induce the same future probability law on the observer-accessible event algebra. If their accessible states differ by trace distance at most ε, then the induced future history laws differ in total variation by at most ε.*

Proof. All future observer-accessible probabilities are computed by applying the same completely positive update maps and the same event-readout maps from the same semantic future-law class to the same accessible algebra and external interface data. Exact equality gives equality of all future event probabilities. In the approximate case, contractivity of trace distance under completely positive trace-preserving maps gives the stated total-variation bound. □

10 Central Records and Measurement

The measurement package is a finite central-record package. Records are exposed by observer-facing patch subfederations.

Let $\mathcal{Z}_{\mathrm{rec}}(t)$ be the commutative algebra generated by the completed, observer-accessible record projectors at cycle t. An event E is a projector in $\mathcal{Z}_{\mathrm{rec}}(t)$. The operational measurement rule is:

$$\Pr(E) = \mathrm{Tr}(\rho E), \qquad \rho \mapsto \frac{E\rho E}{\mathrm{Tr}(\rho E)} \quad \text{when } \Pr(E) > 0.$$

Theorem 10.1 (Federated central-record measurement)**.** *On a fixed-cutoff federated patch carrier, once a completed write/verify slice exposes a finite commutative central record algebra $\mathcal{Z}_{\mathrm{rec}}(t)$, the Born probability and Lüders conditioning rule above define the operational measurement package on the observer-accessible event surface. Re-reading the same completed record event has probability one, conditional on that record event.*

Proof. The theorem is the standard finite-dimensional central-record argument. Because all declared record events commute and belong to the observer-accessible center for the completed slice, they define an ordinary finite classical event algebra. Probabilities are Born traces on that event algebra, and conditioning on an event is the Lüders update. After conditioning on E, the same central projector E is true with probability one. □

11 Edge Sectors and the Casimir Handoff

The edge heat-kernel/Casimir package is a fixed-cutoff theorem package tied to an overlap collar with a finite exposed sector algebra.

At fixed cutoff, let α label a finite set of exposed edge sectors on one declared overlap collar. Suppose the local thermalized edge dynamics has stationary weights

$$\pi_\beta(\alpha) = \frac{d_\alpha e^{-\beta C_2(\alpha)}}{Z(\beta)}.$$

For finite groups, $C_2(\alpha)$ is the declared sector penalty or finite-group Casimir surrogate. For compact groups, the Peter–Weyl lift belongs to the companion D10/compact-gauge handoff instead of hardware evidence.

Theorem 11.1 (Fixed-cutoff edge-sector handoff)**.** *If a declared finite overlap collar has sector labels α, degeneracies d_α, and a local repair/thermalization generator whose detailed-balance stationary law is $\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)}$, then the collar exports the fixed-cutoff Casimir edge law required by the D10 handoff. The compact-group heat-kernel lift is a separate companion-branch step.*

Proof. This is a finite Markov or finite instrument stationary-measure statement on the declared sector algebra. The theorem exports only the finite overlap law. The compact-group lift uses the Peter–Weyl continuation and normalization conventions supplied on the companion compact/D10 surface. □

12 Bell/CHSH Event Surfaces

The Bell package is fixed-cutoff and event-surface-local. A federated carrier may contain two observer-facing wings L and R with commuting setting and outcome records on one compare slice. If the source-specified joint law is the usual two-wing quantum law, the CHSH and Tsirelson statements are carried on that declared event algebra.

Laboratory echosahedral hardware has its own evidence gate for Bell-style claims. The theorem package here is a finite event-algebra statement inside the OPH microphysics surface.

13 Relation to Yang–Mills

The Yang–Mills repair-gap argument belongs to the compact theorem surface. Echosahedral geometry supplies:

1. the fixed-cutoff patch, overlap, record, and repair interface;
2. the finite edge-sector Casimir law on declared collars;
3. the local repair semantics that the compact-gauge branch later reads in controlled quotient form.

The four-dimensional Yang–Mills form, reflection-positive ordinary vacuum branch, controlled continuum extraction, and equality between the Yang–Mills gap and the repair gap belong to the compact/Yang–Mills theorem surface. Hardware evidence may test implementation discipline, but it does not supply the Clay-admissible construction. A public evidence bundle that is used for any continuum or Lorentzian claim must therefore include the multiresolution regulator certificate: factor manifest, presentation-circuit hash, refinement/expectation identity defects, transported-state errors, renormalized-observable tails, and positive-transfer or reflected-Gram receipts. Conventional free-field or lattice-gauge vacuum baselines used for calibration are reference ensembles only. They are not OPH-native vacuum promotion unless the quotient-ensemble selector, source Euclidean slab data, and transfer/reflection-positive reconstruction gate are supplied on the compact theorem surface.

14 Relation to Hadrons and Strong-Binding Execution

First-principles hadron masses require a working OPH strong-binding construction. This microphysics states the record and evidence requirements for such a construction. A construction fit for promotion must emit:

1. Ward-projected hadronic spectral data;
2. provenance tying every spectrum to body, controller, firmware, geometry, and scorebook;
3. finite-volume, continuum, chiral, and production-systematics fields where applicable;
4. exact replay or verification receipts for the pipeline stage being claimed;
5. a non-promotion boundary for surrogate or calibration runs.

Echosahedral or related hardware can be described as a candidate construction family only through public evidence bundles. Without those bundles, it is an architectural target and a design motivation. First-principles hadron theorem promotion requires the evidence listed above.

15 Validation Program

The validation program has two independent tracks.

15.1 Mathematical and Digital Calibration

The octahedral $\mathbb{Z}_2/S_3$ finite-group model is a digital calibration model. It gives exact finite groups, explicit patch covers, controlled defects, frustrated cycles, record writes, repair schedules, and negative controls.

> The calibration model checks the finite interface logic. The primary physical picture is the federated patch-carrier architecture.

15.2 Public Hardware Evidence

A public hardware run may support only the support level its evidence bundle can justify. The minimal claim form is:

> Module set M produced candidate enrichment or reproducible readout signature on benchmark T, under controls C, and exact verifier V accepted the reported hits.

The rejected form is:

> The optical chamber solved the hard problem or proved OPH.

Required gates include dark baseline, low-power sweep, coupling matrix, discharge-timing or MDD trace where applicable, ring-diversity or recurrence-sensitive metrics for toroidal bodies, duplicate-body checks, symmetric-reference or echosahedral shadow checks, exact-verifier receipts, and negative controls against hidden duplicate amplification or host-side filtering.

Material, plasma, and topological-phase claims need the same discipline. The evidence bundle must name the physical quotient, source action or repair ledger, readouts, controls, and promotion gates. A carrier that passes self-reading tests certifies the carrier branch only; it does not by itself certify a material selector, nuclear yield, delivered power, or condensed-matter order.

16 Finite Packet Source Bridge Schema

Cosmology-facing evidence bundles may not infer stress data from scalar source rows. A packet-source bridge exported from the microphysics layer must carry the primitive data needed by the finite covariant parent contract:

$$(\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r).$$

In practical evidence bundles this means:

1. finite cell/face incidence, four-volumes, normals, areas, causal adjacency, and parallel transport maps;
2. metric and tetrad data with the declared $(- + ++)$ convention;
3. packet sectors, local momenta, masses, positive invariant weights, occupations, and mass-shell residuals;
4. signed face fluxes and reaction-channel stoichiometry with transported four-momentum residuals;
5. quotient maps, local-frame covariance checks, and restriction/refinement maps;
6. readout fields for stress moments, variational stress, exchange currents, and gauge-invariant or rest-frame perturbation variables.

The corresponding finite evidence receipt is residual-based. A row labelled "recipient", a raw Newtonian/synchronous gauge comparison, a producer-declared detailed-balance boolean, or a frozen likelihood hash is not a substitute for primitive packet, stress, exchange, causal-response, and refinement evidence.

17 Finite Quotient Ensemble Compatibility

Finite quotient ensemble theorem surface. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any map $N : Q \to Q_{\mathrm{nf}}$, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\# \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_\# \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_\# \mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_\# \mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_\# \widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_*, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound, not just a finite histogram.

Vacuum promotion gate. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \ge 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance is not automatically a primordial curvature spectrum. The map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has a null space unless a radial prior, finite parametric family, or source-stress bridge is declared. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, screen-to-radial-lift, radial-null-space, and forward-projection receipts. Observable CMB promotion further requires frozen source, solver, likelihood, no-data-use, pooled-reducer, and falsification-rule receipts.

Claim tiers and required receipts. Every ensemble-facing run records immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, canonicalizer hash, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler hash, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. The claim tiers are

$$\begin{aligned} E0 &: \text{ seed noise, proposal noise, repair jitter,} \\ E1 &: \text{ conventional reference ensemble,} \\ E2 &: \text{ OPH-native quotient ensemble,} \\ E3 &: \text{ OPH vacuum,} \\ E4 &: \text{ OPH primordial field,} \\ E5 &: \text{ observable cosmological prediction.} \end{aligned}$$

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

18 Conclusion

OPH microphysics is a federation of finite observer patches. The sphere is a regulator and symmetry chart for observer-facing cuts. A_5-icosahedral and E_8-type structure belong to the geometry data and representation-closure data. The echosahedral patch is the reference local interface: a bounded multi-port patch with symmetry, records, readout, repair, and checkpoint data. Toroidal subchannels supply local recurrence and winding-sensitive dynamics. The mathematical exports remain fixed-cutoff patch-net embedding, edge-sector Casimir handoff, central-record measurement, Bell/CHSH event surfaces, and checkpoint/restoration.

The claim discipline is the main point. Hardware can guide the architecture and supply public evidence through hash-stable evidence bundles. The theorem surface stands on finite algebras and declared OPH branch assumptions.

A Evidence Bundle Sketch

A minimal hardware evidence bundle should use a structure like:

```
evidence/hardware/<bundle-id>/
  manifest.json
  README.md
  body/
    mesh_hashes.txt
    photos/
    measurements.csv
  controller/
    firmware_sha256.txt
    wiring_map.csv
  calibration/
    dark_scan.csv
    low_power_sweep.csv
    coupling_matrix.csv
    mdd_trace.csv
```

```
    ring_diversity.csv
  task/
    task.json
    scorebook.json
    candidates.jsonl
    verifier_receipts.jsonl
  controls/
    shuffle_replay.jsonl
    abba_controls.csv
    negative_controls.md
  claim.md
```

The manifest should state the strongest allowed claim and the non-claims. The paper may cite the bundle only at that claim level.

B Digital Calibration Compatibility

The octahedral $\mathbb{Z}_2/S_3$ build has the following reading rule:

1. it validates finite patch/overlap/record/repair bookkeeping;
2. it tests frustrated-cycle and defect behavior in an exact digital setting;
3. it calibrates the edge-sector law on a finite declared interface;
4. it supplies calibration data for the interface layer;
5. it leaves the physical carrier role with the federated echosahedral patch architecture.

References

[1] B. Müller, K. Xue, K. A. Anirudha, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics.*

[2] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, and M. A. Visser, *Recovering Relativity and the Standard Model from Observer Overlap Consistency.*

[3] B. Müller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, M. A. Visser, K. A. Anirudha, and J. Hill, *Observers Are All You Need.*

[4] B. Müller, A. Osika, M. Poneder, K. Xue, and M. A. Visser, *Deriving the Particle Zoo from Observer Consistency.*

[5] B. Müller, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics: A Support-Visible OPH Route to the Clay Problem.*